



Roll No.:

Total Printed Pages: 2

AM100201

CT- I (March 26)

Subject- Applied Maths-II

B. TECH - II SEM (P2 - GROUP)

Maximum Marks: 40

Min Marks- 14

Duration: 2 hour

Note:-

- 1) All questions are compulsory.

Section 1

Q. (1) Objective Question:

(4x1=4)

1. The infinite series $\sum 1/n^2$ is convergent or divergent and why?

2. What is the necessary condition for an infinite series, $\sum u_n$ to be convergent?

3. State intermediate value property.

4. State D'moivre's theorem.

Or

Show that the Sequence:

$\frac{2^2}{3!}, \frac{3^2}{4!}, \frac{4^2}{5!}, \frac{5^2}{6!}, \dots$ is convergent.

Q. (2) Short Answer Type Question:

(3x4=12)

1. Test for convergence: $\sum (1 + \frac{1}{n})^{-n}$.

2. Show that the equation; $x^7 - 3x^4 + 2x^2 - 1 = 0$ has at least four imaginary roots.

3. Separate: $\tan(x + i y)$ into real and imaginary parts.

Or

JUNWANI, BHILAI - 490 020 (CHHATTISGARH), INDIA

Phone : 0788-4088888, 2298605, Email : ssfc@ssfc.ac.in, Website : www.ssfc.ac.in

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Solve: $4x^4 - 20x^3 + 33x^2 - 20x + 4 = 0$.

Section 2

Q. (3) Descriptive Answer Type Question: (Solve any 3) (3x8=24)

1. Test the convergence of the series;

$$\sum (\sqrt{n^4 + 1} - \sqrt{n^4 - 1}).$$

2. Solve: $6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$.

3. If $x_r = \frac{\pi}{2^r} + i \sin \frac{\pi}{2^r}$.

Prove that: $x_1 x_2 \dots \infty = -1$.

Or

Test the convergence of the series;

$$1 + \frac{x}{2} + \frac{1.3}{2.4} x^2 + \frac{1.3.5}{2.4.6} x^3 + \frac{1.3.5.7}{2.4.6.8} x^4 + \dots$$

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CT- II (April 2026)

Subject- Applied Maths-II

B. TECH – II SEM (P2 - GROUP)

Maximum Marks: 40

Min Marks- 14

Duration: 2 hour

Note:-

Section-A is compulsory containing 4 objective type questions of 4 marks and 03 short answer type questions, carrying 04 marks for each, total of 12 marks.

Section-B containing 06 descriptive type questions, two from each unit with 50% internal choice, carrying 08 marks for each, total of 24 marks.

Section - A

Q. (1) Objective Question:

(1x4=4)

(a). The Evaluation of $\Gamma(-7/2)$ is equal to:

(i) $\frac{8}{105}$ (ii) $\frac{16}{105}\sqrt{\pi}$ (iii) $\frac{8}{105}\sqrt{2\pi}$ (iv) None of these.

(b). If $\text{div}\vec{V} = 0$, then the vector is.....

(c). State Green's theorem.

(d). Define positive correlation with an example.

Q. (2) Short Answer Type Question:

(4x3=12)

(a). Prove that: $\beta(m, n) = \beta(n, m)$.

(b). Prove that: $\text{div}\vec{r} = 3$.

(c). Prove that the G. M. of two regression coefficient is equal to the correlation coefficient.



Section - B

Q. (3) Descriptive Answer Type Question:

(8x3=24)

1. Evaluate by changing the order of integration: $I = \int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dx dy$.
2. Find the area between the parabola, $y^2 = 4ax$ and $x^2 = 4ay$ by double integration.
3. Evaluate the line integrals $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed by the lines $x = \pm 1, y = \pm 1$.
4. Prove that: $\nabla^2 f(\vec{r}) = f''(\vec{r}) + \frac{2}{r} f'(\vec{r})$.
5. Fit a second degree parabola from the following data:
x: 1.0 1.5 2.0 2.5 3.0 3.5 4.0
y: 1.1 1.3 1.6 2.0 2.7 3.4 4.1
6. Calculate the coefficient of correlation from the following data:
x: 105 104 102 101 100 99 98 96 93 92
y: 101 103 100 98 95 96 104 92 97 94.